

Ali Haydar Sucu

✉ alihaydarsucu@gmail.com |  linkedin.com/in/ali-haydar-sucu
 github.com/alihaydarsucu |  alihaydarsucu.github.io
📍 Ankara, Turkey

Education

Ankara Yıldırım Beyazıt University

Ankara, Turkey

Bachelor of Science in Computer Engineering

Sep 2023 – Jun 2028 (Expected)

- Current GPA: 3.36/4.00
- Activities: AYBÜ Hezarfen Student Club, Developer Students Community, IEEE Student Branch - AYBU

Tevfik İleri Anatolian High School

Ankara, Turkey

High School Diploma

Sep 2019 – Jun 2023

- Graduated with honors: 96.43/100

Professional Experience

AI Developer | Part-Time Student

Jun 2025 – Present

AYBU ÖĞMER

Ankara, Turkey

- Led development of an intelligent document-interactive AI agent using Llama 4 LLM with web interface published for university-wide access
- Implemented core features: document processing (PDF, DOCX, TXT), question answering, text summarization, and Turkish speech-to-text/text-to-speech capabilities
- Designed system architecture with real-time audio streaming and document embedding for knowledge retrieval

Embedded Software Developer

Oct 2024 – Aug 2025

PUSULA USV

Ankara, Turkey

- Developed and optimized autonomous navigation and control algorithms for embedded systems on NVIDIA Jetson platform using Python.
- Engineered systemd services and real-time system integration for launching multi-threaded ROS 2 nodes handling sensor data streams (Lidar, ZED Camera) and autonomous control.
- Integrated and debugged multiple embedded sensors: A2M12 Lidar sensor fusion, ZED 2 stereo camera processing, Pixhawk flight controller motor commands, and GPS.
- Collaborated with Electronics team on hardware-software co-design, conducting extensive real-world testing, debugging sensor communication protocols, and optimizing system performance

Avionics & Embedded Systems Team Member

Sep 2023 – Jun 2024

AYBÜ HEZARFEN Club | Gökçen Rocket Project

Ankara, Turkey

- Conducted detailed sensor research and selection for rocket avionics systems.
- Contributed to embedded flight control board design and development.
- Performed embedded system integration and testing of avionics hardware components, serial communication debugging, and sensor calibration
- Developed system specifications documentation and conducted initial flight avionics validation for Teknofest's rocket competition

Featured Projects

STM32F103RB Embedded Systems Learning | C/C++, STM32CubeIDE, HAL, ARM Architecture 2024 – 2025

- GitHub: github.com/alihaydarsucu/STM32-Learning
- Embedded systems coursework on NUCLEO-F103RB MCU platform covering peripherals: GPIO configuration, UART serial communication (115200 baud), PWM generation, timer operations, and HAL library utilization
- Developed low-level register-level programming skills and STM32 microcontroller architecture understanding through hands-on firmware development

FocusPath - Focus Tracking & Analysis C++, Qt 5/6, X11 APIs, SVG Rendering, Linux		2025
• GitHub: github.com/alihaydarsucu/FocusPath • Developed low-level system monitoring application capturing X11 window events at 200ms intervals for focus tracking and distraction analysis on Linux desktop • Engineered real-time window tracking system using direct X11 APIs, JSONL event logging, and advanced system integration for behavioral analytics • Implemented SVG chart generation pipeline rendering usage statistics, timeline visualizations, and attention metrics for productivity analysis • Designed cross-platform C++ architecture demonstrating systems programming expertise, direct OS-level API interaction, and performance-critical event handling		
Pusula Ground Control Station Python, FastAPI, Real-Time Telemetry, System Integration		Jul – Aug 2025
• GitHub: github.com/abdullah-aksoy/Pusula-GCS • Designed and developed embedded-friendly GCS software for autonomous USV platform with real-time telemetry processing at high update rates • Implemented command-and-control interface for embedded system operations including ARM/DISARM commands, mode switching, algorithm control, and real-time vehicle state monitoring • Integrated mission planning with geographic data processing and live tracking visualization of autonomous vehicle operations		
Personal Portfolio With CV Generation LaTeX, GitHub Actions, CI/CD, System Automation		2024
• GitHub: github.com/alihaydarsucu/alihaydarsucu.github.io • Engineered automated CV generation pipeline using LaTeX and GitHub Actions for continuous ATS-compatible document generation • Designed system automation workflow demonstrating CI/CD understanding and proficiency with build automation tools		

Technical Skills

Languages: C/C++, Python, Bash/Shell, Java, JavaScript
Embedded Systems: STM32, ARM Architecture, ROS 2, Jetson, Sensor Integration
Tools: Git, Linux, Docker, STM32CubeIDE, ST-LINK Debugger
Languages: Turkish (Native), English (Professional)

Certifications & Achievements

Teknofest Success Certificate		Nov 2025
<i>Teknofest</i>		
• Ranked 10th among 284 teams in Mavi Vatan competition		
Career Module Certificate		July 2025
<i>MSMB - Mimar Sinan Mühendisler Birliği</i>		
• Professional career module certification		
Data Science Bootcamp Certification		Dec 2023
<i>Developer Students Community</i>		
• Completed intensive bootcamp covering Python programming and data processing		